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Q1. What are thin-film and thick-film technologies?

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Through e-mail

A1. *Thin-film technology.* Thin-film deposition is the technique for depositing a thin film of any material onto a substrate or onto previously deposited layers. Most deposition techniques allow layer thickness to be controlled within a few tens of nanometres. However, some deposition techniques, such as molecular beam epitaxy, allow single layers of atoms to be deposited at a time.

Thin films are made of thin material layers ranging from fractions of a nanometre to several micrometres in thickness. Electronic semiconductor devices and optical coatings are the main applications of thin-film construction. Ceramic thin films are widely used for protection of various materials against corrosion, oxidation and wear.

Thin-film deposition techniques fall into two broad categories, depending on whether the process is primarily chemical or physical.

Chemical solution deposition uses a liquid precursor, usually a solution of organometallic powders dissolved in an organic solvent.

Chemical vapour deposition (CVD) generally uses a gas-phase precursor, often a halide or hydride of the element to be deposited. In the case of MOCVD, an organometallic gas is used. Commercial techniques often use very low pressures of precursor gas.

Plasma-enhanced CVD uses an ionised vapour, or plasma, as the precursor. It uses electromagnetic means, e.g., electric current and microwave excitation, rather than a chemical reaction, to produce plasma.

Physical deposition uses mechanical or thermodynamic means to produce a thin film of solid. Since most engineering materials are held together by relatively high energies, and chemical reactions are not used to store these energies, commercial physical deposition systems tend to require a low-pressure vapour environment to

function properly; most can be classified as physical vapour deposition.

Examples of physical deposition systems include thermal evaporators, electron beam evaporators, sputtering system, pulsed laser deposition systems and cathodic arc deposition systems.

Thin-film technologies are being developed as a means of substantially reducing the cost of photovoltaic (PV) systems. The rationale for this is that thin-film modules are cheaper to manufacture owing to their reduced material costs, energy costs, handling costs and capital costs. These are especially used in printed electronic (roll-to-roll) processes.

Research is being done on a new class of thin-film inorganic oxide materials, called amorphous heavy-metal cation multicomponent oxides, which could be used to make transparent transistors that are inexpensive, stable and environmentally benign. Ferromagnetic thin films are being researched for use as computer memories. Thin-films are also being used to produce thin-film batteries.

Thick-film technology. It is used to produce such electronic devices as surface-mount devices, hybrid integrated circuits and sensors. Manufacturing of these devices entails deposition of several successive layers onto an electrically insulating substrate using a screen-printing process.

Typically, the thick-film process technology consists of the following stages: ink preparation, screen-printing, drying/curing, firing, and separation of elements and integration of devices.

Ink preparation. Inks for electrodes, terminals, resistors, dielectric layers, etc are commonly prepared by mixing the metal or ceramic powders required with an organic vehicle to produce a paste for screen-printing. To achieve homogeneous ink, the mixed components of the ink may be passed through a three-roll mill. Alternatively, ready-made inks may be obtained from one of the many companies offering prod-